

Practice Questions Part 2

1. A randomized experiment ensures that a treatment group and a control group are _____ each other on pretreatment characteristics.

- a. different from
- ☒ b. comparable to
- c. neither a nor b is correct; randomized experiments are only connected to posttreatment outcomes.

2. The Washington DC Gun Study is an example of a randomized experiment.

- a. true
- ☒ b. false

3. If a city's police department knows about 100 household burglaries in a year's time and we believe the chance that a household burglary is reported to the police is somewhere between 35% and 63%, then what would be a reasonable estimated range of actual household burglaries?

$100/0.35 = 285.714$ With these reporting rates, the estimated
 $100/0.63 = 158.73$ range of household burglaries would be [159,286].

4. Suppose we have 2 states. State 1 has a population of 1 million people and State 2 has a population of 2 million people. Each state experiences 300 homicides in a year's time. Calculate the number of homicides per 100,000 population for each state. Be sure to show your calculations.

$(300/1,000,000) \times 100,000 = 30$ State 1's rate is 30 homicides per 100,000
 $(300/2,000,000) \times 100,000 = 15$ persons while state 2's rate is 15 homicides
per 100,000 persons.

5. If the local police department begins a new school resource officer (SRO) program in 1/2 of the city's schools and we find that juvenile arrest rates decrease after the program begins among the kids who attend those schools, have we demonstrated that the SRO program is effective at reducing juvenile arrest rates? Why or why not?

No because, establishing that a comes before b does not prove that a causes b. We would need a convincing control group.

6. The Washington DC Gun Study examined which of the following outcomes?

- a. robberies
- b. aggravated assault
- c. homicides
- d. suicides
- e. both a and c are correct
- ☒ f. both c and d are correct

7. Which of the following studies represented a collaboration between the local police and the researchers conducting the study?

- a. the residential burglary study (my paper)
- ☒ b. the Washington DC Gun Study
- c. the Kansas City Gun Experiment
- d. none of the above are correct